

About the Current OSPO VSIP Training Curriculum

The Georgia Tech OSPO Virtual Summer Internship Program (VSIP) pairs undergraduate and graduate students with open-source projects for a 10-week summer project, under the guidance of a mentor. During this program, students will undergo training sessions to learn fundamental skills for open-source development.

The current OSPO VSIP training curriculum contains the following modules:

1. Intro to Git
2. Forking repositories
3. Testing code
4. CI/CD pipelines
5. Code review
6. Governance

Project Overview

Motivation: Currently, the Georgia Tech OSPO OSS MOOC training consists of a set of Jupyter Notebooks which contains notes and some code cells that can be run by students. During the OSPO VSIP program, every week an OSPO mentor leads a workshop running through the Jupyter Notebook. There are three main problems with the current system:

1. The notebooks by themselves are not interactive.
2. Some parts of the notebook are not as fleshed out which means that a mentor is needed to guide the students through what to learn.
3. The notebooks in conceptual-heavy lessons do not contain any thing to test student's knowledge on things they have learnt in the section.

To address the issue, a new OSS MOOC training platform is developed using Jupyter Hub and nbgrader, which enables automated grading.

Goals and Milestones

There are three main goals in this project:

1. Increase the interactivity of existing lessons by deploying the platform to a Jupyter Hub server and implementing tasks that can be automatically graded using nbgrader.
2. Systematize the setup and contribution process for future contributors by setting up formal documentation that is written using Sphinx and deployed on ReadTheDocs.
3. Design and add new interactive content (e.g., quizzes, code cells) to existing Jupyter Notebooks.

To achieve these three goals throughout the summer, my mentor and I designated four major milestones:

1. **Consolidate developer documentation.**
2. **Implement Jupyter Hub and nbgrader workflow.**
3. **Unify pre-existing codebases.**
4. **Add additional content.**

Open-Source Outcomes

- It is easy to contribute to open-source projects, even large, influential projects!
- Obtained a firm grasp of normal contributor etiquette and procedures.
- Learnt more about what CI/CD pipelines are.
- Learnt that there are a variety of ways for an open-source project to be governed, and that there may be different reasons as to why certain projects are governed a certain way.

Future Work

1. Continue to work with the CoC IT department to deploy a Jupyter Hub server on a RHEL VM.
2. Clean up documentation of setup and installation process.
3. Define a firm ruleset of contribution procedures to enable friendlier and more efficient open-source contribution.
4. Implement a CI/CD pipeline which enables automated checks prior to committing code.

Highlights and Accomplishments

- Completed milestone 1 by consolidating existing developer documents from previous contributors into a Sphinx docs website, which is deployed using ReadTheDocs.
- Learnt and setup nbgrader on a local Jupyter Hub environment.
- Added new interactive content to all existing Jupyter Notebooks and consolidated existing code into one nbgrader source code directory.

